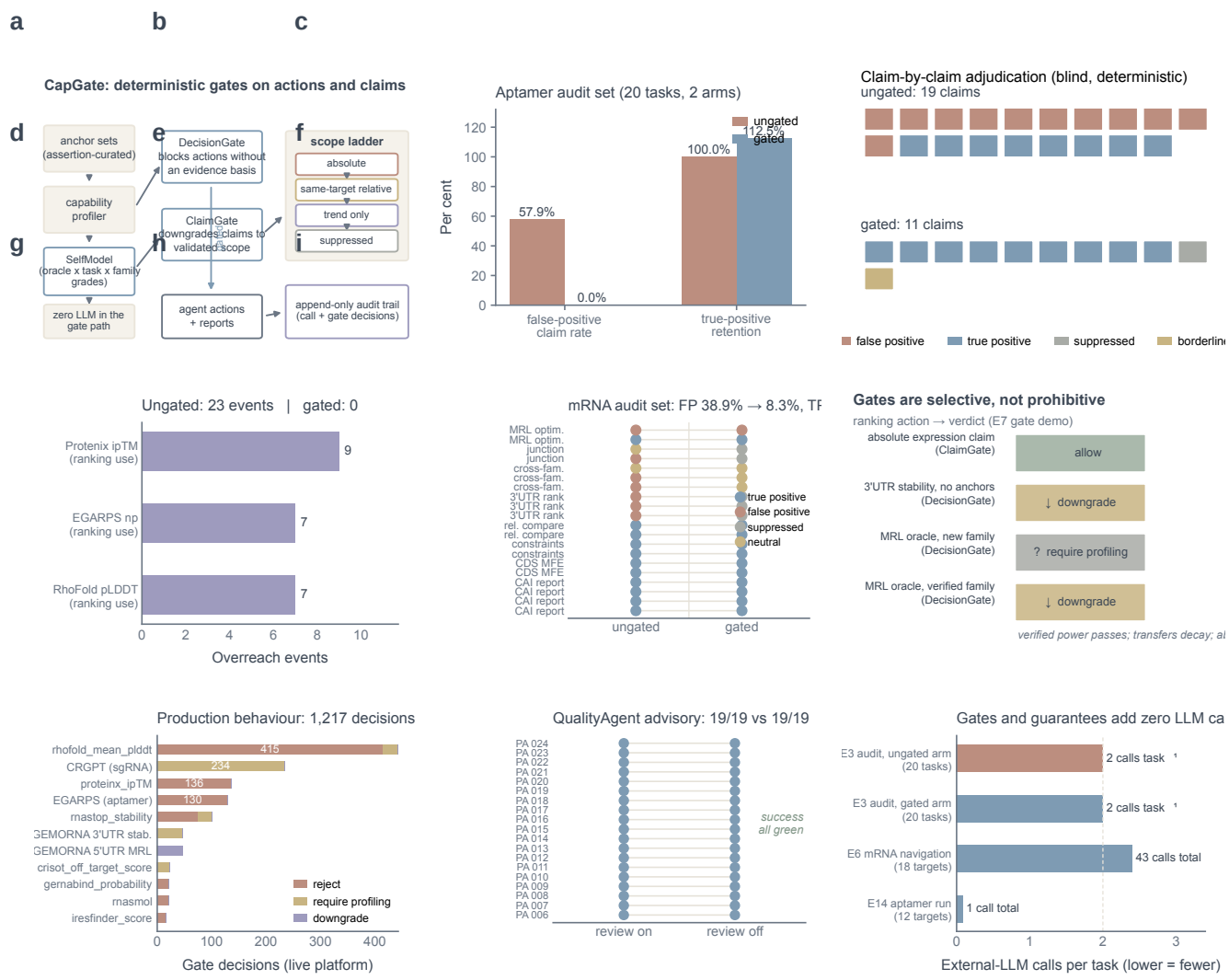

a
b
c
CapGate: deterministic gates on actions and claims
d
anchor sets (assertion-curated)
capability profiler
g
SelfModel (oracle x task x family grades)
zero LLM in the gate path
e
DecisionGate blocks actions without an evidence basis
ClaimGate downgrades claims to validated scope
h
agent actions + reports
append-only audit trail (call + gate decisions)
f
scope ladder
absolute
same-target relative
trend only
i
suppressed
Aptamer audit set (20 tasks, 2 arms)
Per cent
ungated
gated
57.9%
0.0%
100.0%
112.5%
false-positive claim rate
true-positive retention
Claim-by-claim adjudication (blind, deterministic)
ungated: 19 claims
gated: 11 claims
false positive
true positive
suppressed
borderline
Ungated: 23 events | gated: 0
Protenix ipTM (ranking use)
EGARPS np (ranking use)
RhoFold pLDDT (ranking use)
Overreach events
mRNA audit set: FP 38.9% → 8.3%, TP
true positive
false positive
suppressed
neutral
ungated
gated
Gates are selective, not prohibitive
ranking action → verdict (E7 gate demo)
absolute expression claim (ClaimGate)
allow
3'UTR stability, no anchors (DecisionGate)
↓ downgrade
MRL oracle, new family (DecisionGate)
? require profiling
MRL oracle, verified family (DecisionGate)
↓ downgrade
verified power passes; transfers decay; al
Production behaviour: 1,217 decisions
rhofold_mean_plddt
CRGPT (sgRNA)
proteinx_ipTM
EGARPS (aptamer)
rnastop_stability
GEMORNA 3'UTR stab.
GEMORNA 5'UTR MRL
crisot_off_target_score
gernabind_probability
rnasmol
iresfinder_score
reject
require profiling
downgrade
Gate decisions (live platform)
QualityAgent advisory: 19/19 vs 19/19
success all green
review on
review off
Gates and guarantees add zero LLM ca
E3 audit, ungated arm (20 tasks)
2 calls task
E3 audit, gated arm (20 tasks)
2 calls task
E6 mRNA navigation (18 targets)
43 calls total
E14 aptamer run (12 targets)
1 call total
External-LLM calls per task (lower = fewer)